

UE21MA141B - Linear Algebra and Its applications

Unit 5 - Singular Value Decomposition

Class work Problems

1) Write the symmetric matrix which corresponds to the following quadratic forms :-

i) $Q(x) = 2x_1^2 + 7x_2^2 + 2x_3^2 - 6x_1x_2 + 5x_2x_3$

ii) $Q(x) = 3x_1^2 + 6x_2^2 - 7x_3^2 - \pi x_4^2 - \sqrt{2}x_1x_2 + 4x_1x_3 - 2x_2x_3 - \pi x_3x_4 + 10x_2x_4 - \frac{1}{2}x_1x_4$

iii) $Q(x) = 10x_1^2 - 6x_1x_2 - 3x_2^2$

solution :- i) $A = \begin{bmatrix} 2 & -3 & 0 \\ -3 & 7 & 5/2 \\ 0 & 5/2 & 2 \end{bmatrix}$

ii) $A = \begin{bmatrix} 3 & -\sqrt{2}/2 & 2 & -1/4 \\ -\sqrt{2}/2 & 6 & -1 & 5 \\ 2 & -1 & -7 & -\pi/2 \\ -1/4 & 5 & -\pi/2 & -\pi \end{bmatrix}$

iii) $A = \begin{bmatrix} 10 & -3 \\ -3 & -3 \end{bmatrix}$

2) compute the quadratic form $x^T A x$ for

$A = \begin{pmatrix} 4 & 3 & 0 \\ 3 & 2 & 1 \\ 0 & 1 & 1 \end{pmatrix}$ and a) $x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$ b) $x = \begin{pmatrix} 2 \\ -1 \\ 5 \end{pmatrix}$

c) $x = \begin{pmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{pmatrix}$

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solution: a)
$$\begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 4 & 3 & 0 \\ 3 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 4x_1 + 3x_2 \\ 3x_1 + 2x_2 + x_3 \\ x_2 + x_3 \end{bmatrix}$$

$$= x_1(4x_1 + 3x_2) + x_2(3x_1 + 2x_2 + x_3) + x_3(x_2 + x_3)$$

$$= 4x_1^2 + 3x_1x_2 + 3x_1x_2 + 2x_2^2 + x_2x_3 + x_2x_3 + x_3^2$$

$$= 4x_1^2 + 2x_2^2 + x_3^2 + 6x_1x_2 + 2x_2x_3$$

b)
$$\begin{bmatrix} 2 & -1 & 5 \end{bmatrix} \begin{bmatrix} 4 & 3 & 0 \\ 3 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & -1 & 5 \end{bmatrix} \begin{bmatrix} 8-3 \\ 6-2+5 \\ -1+5 \end{bmatrix} = \begin{bmatrix} 2 & -1 & 5 \end{bmatrix} \begin{bmatrix} 5 \\ 9 \\ 4 \end{bmatrix} = 10 - 9 + 20$$

$$= 21$$

c)
$$\begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \end{bmatrix} \begin{bmatrix} 4 & 3 & 0 \\ 3 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$$

$$= \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \end{bmatrix} \begin{bmatrix} 7/\sqrt{3} \\ 6/\sqrt{3} \\ 2/\sqrt{3} \end{bmatrix} = 7/3 + 6/3 + 2/3 = \frac{15}{3} = 5$$

3) Decide against the positive definiteness of these matrices and write the corresponding quadratic form $f = x^T A x$?

i) $\begin{pmatrix} 1 & 3 \\ 3 & 5 \end{pmatrix}$ ii) $\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$ iii) $\begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{pmatrix}$ iv) $\begin{pmatrix} 0 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{pmatrix}^2$

Solution :- i) $\begin{bmatrix} 1 & 3 \\ 3 & 5 \end{bmatrix}$ $A_1 = 1$; $|A_2| = 5 - 9 = -4 < 0$

∴ Matrix is not positive definite

ii) $B = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$; $|B_1| = 1$; $|B_2| = 1 - 1 = 0$

Eigen value test :- $\begin{vmatrix} 1-\lambda & -1 \\ -1 & 1-\lambda \end{vmatrix} = 0$

$$(1-\lambda)^2 - 1 = 0 \Rightarrow \lambda^2 + 1 - 2\lambda - 1 = 0$$

$$\lambda^2 - 2\lambda = 0$$

$$\lambda(\lambda - 2) = 0$$

$$\lambda = 0 \text{ \& } \lambda = 2$$

∴ Eigen values are positive.

Pivot test :-

$$\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \xrightarrow{R_2 + R_1} \begin{bmatrix} \textcircled{1} & -1 \\ 0 & \textcircled{0} \end{bmatrix}$$

⇒ Pivots are positive. All the three tests are satisfied. ∴ B is positive definite.

iii) $\begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{pmatrix}$

Refer lectures notes - Page 23

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$$\text{iv) } \begin{pmatrix} 0 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{pmatrix}^2$$

Refer Page 15 of lecture notes

4) For what a & b are the matrices A and B have all $\lambda > 0$ and are therefore positive definite?

$$A = \begin{pmatrix} a & 2 & 2 \\ 2 & a & 2 \\ 2 & 2 & a \end{pmatrix} \quad B = \begin{bmatrix} 1 & 2 & 4 \\ 2 & b & 8 \\ 4 & 8 & 7 \end{bmatrix}$$

Solution :- Refer Page 17 of lecture notes

9) Find the covariance matrix for $a_1 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$ $a_2 = \begin{pmatrix} 4 \\ 2 \\ 13 \end{pmatrix}$
 $a_3 = \begin{pmatrix} 7 \\ 8 \\ 1 \end{pmatrix}$ $a_4 = \begin{pmatrix} 8 \\ 4 \\ 5 \end{pmatrix}$

Solution :- Refer Page 45 of lecture notes

10) The following table lists the weights and heights of 5 boys :-

Boy	#1	#2	#3	#4	#5
Weight	120	125	125	135	145
Height	61	60	64	68	72

Find the covariance matrix for the above data?

Solution :- Page 46 of lecture notes

5) Compute $A^T A$ and $A \cdot A^T$ and their eigen values and unit eigen vectors, for $A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}_{2 \times 3}$. Verify if $A = U \Sigma V^T$?

Solution :- $A^T \cdot A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}_{3 \times 2} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}_{2 \times 3} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix}_{3 \times 3}$

$$A \cdot A^T = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}_{2 \times 2}$$

To find eigen values :- $\begin{vmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{vmatrix} = 0$

$$(\lambda-2)^2 - 1 = 0 \Rightarrow \lambda^2 - 4\lambda + 3 = 0$$

$$\lambda^2 - 4\lambda + 3 = 0$$

$$\lambda^2 - 3\lambda - \lambda + 3 = 0$$

$$\lambda(\lambda-3) - 1(\lambda-3) = 0$$

$$(\lambda-3)(\lambda-1) = 0$$

$$\boxed{\lambda = 1 \text{ or } \lambda = 3}$$

Eigen values are 1 & 3

To find eigen values :-

$$(2-\lambda)x + y = 0$$

$$x + (2-\lambda)y = 0$$

for $\lambda = 1$

$$x + y = 0$$

$$x + y = 0$$

$$x = -y$$

$$\text{So } x_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

for $\lambda = 3$

$$-x + y = 0$$

$$x - y = 0$$

$$x = y$$

$$\text{So } x_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$u_1 = \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

$$\Rightarrow U = \begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$u_2 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

⑥

$$V_1^T = \frac{U_1^T \cdot A}{\sigma_1} = \frac{1 \times 2}{1} \begin{bmatrix} -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}_{2 \times 3} = \begin{bmatrix} -1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix}$$

$$V_1 = \begin{bmatrix} -1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}$$

$$V_2^T = \frac{U_2^T \cdot A}{\sigma_2} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1/\sqrt{2} & \sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

$$V_2 = \begin{bmatrix} 1/\sqrt{6} \\ \sqrt{2/3} \\ 1/\sqrt{6} \end{bmatrix}$$

V_3 should be orthogonal to both V_1 and V_2 .

$$V_3 = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$-\frac{x}{\sqrt{2}} + \frac{z}{\sqrt{2}} = 0; \quad \frac{x}{\sqrt{6}} + \frac{\sqrt{2}}{\sqrt{3}} y + \frac{1}{\sqrt{6}} z = 0$$

$$-x = -z$$

$x = z$

$$\frac{x}{\sqrt{6}} + \frac{\sqrt{2}}{\sqrt{3}} y + \frac{1}{\sqrt{6}} z = 0$$

$$\frac{z}{\sqrt{6}} + \frac{\sqrt{2}}{\sqrt{3}} y + \frac{1}{\sqrt{6}} z = 0$$

$$\frac{\sqrt{2}}{\sqrt{3}} y = \left(-\frac{1}{\sqrt{6}} - \frac{1}{\sqrt{6}} \right) z$$

$$\frac{\sqrt{2}}{\sqrt{3}} y = \frac{-\sqrt{2}}{\sqrt{6}} z$$

$$y = \frac{-\sqrt{3}}{\sqrt{6}} z$$

$y = -\frac{1}{2} z$

$$\therefore V_3 = \begin{pmatrix} z \\ -z/2 \\ z \end{pmatrix}$$

$$\frac{-z}{\sqrt{2}} + \frac{z}{\sqrt{2}} = 0$$

$$\frac{-z}{\sqrt{2}} = -\frac{z}{\sqrt{2}}$$

$$\sqrt{z^2 + \frac{z^2}{4} + z^2} = 1 \quad \left[\begin{array}{l} \text{Magnitude or norm of} \\ \text{column vector should be 1} \end{array} \right]$$

$$\frac{2z^2 + \frac{z^2}{4}}{4} = 1$$

$$\frac{8z^2 + z^2}{4} = 4$$

$$9z^2 = 4 \Rightarrow z^2 = 4/9 \Rightarrow \boxed{z = 2/3}$$

$$\therefore V_3 = \begin{bmatrix} 2/3 \\ -1/3 \\ 2/3 \end{bmatrix} \Rightarrow V = \begin{bmatrix} -1/\sqrt{2} & 1/\sqrt{6} & 2/3 \\ 0 & \sqrt{2/3} & -1/3 \\ 1/\sqrt{2} & 1/\sqrt{6} & 2/3 \end{bmatrix}$$

$$\therefore A = U \Sigma V^T = \begin{bmatrix} -1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sqrt{3} & 0 \end{bmatrix} \begin{bmatrix} -1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 1/\sqrt{6} & \sqrt{2/3} & 1/\sqrt{6} \\ 2/3 & -1/3 & 2/3 \end{bmatrix}$$

$2 \times 3 \quad \quad 3 \times 3$

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} -1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} -1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 1/\sqrt{2} & \sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

$2 \times 2 \quad \quad 2 \times 3$

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$\therefore A$ is decomposed as $U \Sigma V^T$

8)

Find SVD of the following matrices :-

$$i) \begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix} \quad ii) \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix} \quad iii) \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{bmatrix}$$

Solution :-

$$A = \begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix}_{2 \times 2}$$

$$A \cdot A^T = \begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix} \begin{bmatrix} 4 & -3 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 32 & 0 \\ 0 & 18 \end{bmatrix}$$

To find the eigen values of $A \cdot A^T$:-

$$\begin{vmatrix} 32 - \lambda & 0 \\ 0 & 18 - \lambda \end{vmatrix} = 0; \quad (32 - \lambda)(18 - \lambda) = 0$$

$$576 - 32\lambda - 18\lambda + \lambda^2 = 0$$

$$\lambda^2 - 32\lambda - 18\lambda + 576 = 0$$

$$\lambda(\lambda - 32) - 18(\lambda - 32) = 0$$

$$(\lambda - 32)(\lambda - 18) = 0$$

$$\boxed{\lambda = 32 \text{ and } \lambda = 18}$$

To find eigen vectors :-

$$(32 - \lambda)x = 0$$

$$(18 - \lambda)y = 0$$

$$\text{for } \lambda = 32$$

$$0 \cdot x = 0$$

$$-14y = 0$$

$$0 = 0; x \neq 0$$

$$\Rightarrow x = 1$$

$$-14 \neq 0 \Rightarrow y = 0$$

$$\therefore x_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$u_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\text{for } \lambda = 18$$

$$14x = 0$$

$$0 \cdot y = 0$$

$$14 \neq 0 \Rightarrow x = 0$$

$$0 = 0 \Rightarrow y = 1$$

$$x_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$u_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\therefore U = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} 4\sqrt{2} & 0 \\ 0 & 3\sqrt{2} \end{bmatrix}$$

$$0 \quad 3\sqrt{2}$$

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$$v_1^T = \frac{u_1^T \cdot A}{\sigma_1} = \frac{1}{4\sqrt{2}} \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix}$$

$$= \frac{1}{4\sqrt{2}} \begin{bmatrix} 4 & 4 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\Rightarrow v_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$v_2 = \frac{u_2^T \cdot A}{\sigma_2} = \frac{1}{3\sqrt{2}} \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix}$$

$$= \frac{1}{3\sqrt{2}} \begin{bmatrix} -3 & 3 \end{bmatrix} = \begin{bmatrix} \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\Rightarrow v_2 = \begin{bmatrix} \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\therefore V = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\therefore A = U \Sigma V^T$$

$$\begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 4\sqrt{2} & 0 \\ 0 & 3\sqrt{2} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix} = \begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix}$$

$$\therefore A = U \Sigma V^T$$

10) ii) $A = \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix}_{2 \times 3}$

Solution :- $A \cdot A^T = \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix} * \begin{bmatrix} 4 & 8 \\ 11 & 7 \\ 14 & -2 \end{bmatrix}$

$$= \begin{bmatrix} 333 & 81 \\ 81 & 117 \end{bmatrix}$$

$$\begin{vmatrix} 333-\lambda & 81 \\ 81 & 117-\lambda \end{vmatrix} = 0 \Rightarrow (333-\lambda)(117-\lambda) - 81^2 = 0$$

$$38961 - 333\lambda - 117\lambda + \lambda^2 - 6561 = 0$$

To find the eigen vectors $\Rightarrow$

$$(333-\lambda)x + 81y = 0$$

$$81x + (117-\lambda)y = 0$$

for $\lambda = 360$

$$-27x + 81y = 0$$

$$81x - 243y = 0$$

$$81x = 243y$$

$$\boxed{x = 3y}$$

let $y = 1$

$$\Rightarrow x = 3$$

$$\therefore x_1 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

$$u_1 = \frac{1}{\sqrt{10}} \begin{pmatrix} 3 \\ 1 \end{pmatrix} ; u_2 = \frac{1}{\sqrt{10}} \begin{pmatrix} 1 \\ -3 \end{pmatrix}$$

$$\therefore U = \frac{1}{\sqrt{10}} \begin{bmatrix} 3 & 1 \\ 1 & -3 \end{bmatrix}$$

$$\lambda^2 - 450\lambda + 32,400 = 0$$

$$\lambda^2 - 360\lambda - 90\lambda + 32,400 = 0$$

$$\lambda(\lambda - 360) - 90(\lambda - 360) = 0$$

$$(\lambda - 360)(\lambda - 90) = 0$$

$$\boxed{\lambda = 360 \text{ or } \lambda = 90}$$

for $\lambda = 90$

$$243x + 81y = 0$$

$$81x + 27y = 0$$

$$81x = -27y$$

$$x = -\frac{1}{3}y$$

let $y = -3$

$$x = 1$$

$$\therefore x_2 = \begin{pmatrix} 1 \\ -3 \end{pmatrix}$$

$$v_1^T = \frac{u_1^T \cdot A}{\sigma_1} = \frac{1}{6\sqrt{10}} \begin{bmatrix} 3 & 1 \\ \sqrt{10} & \sqrt{10} \end{bmatrix} \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix}$$

$1 \times 2 \qquad \qquad \qquad 2 \times 3$

$$= \frac{1}{6\sqrt{10}} \begin{bmatrix} 20 & 40 & 40 \\ \sqrt{10} & \sqrt{10} & \sqrt{10} \end{bmatrix}$$

$$v_1^T = \begin{bmatrix} \frac{20}{60} & \frac{40}{60} & \frac{40}{60} \end{bmatrix}$$

$$v_1 = \begin{bmatrix} 1/3 \\ 2/3 \\ 2/3 \end{bmatrix}$$

$$v_2^T = \frac{u_2^T \cdot A}{\sigma_2} = \frac{1}{3\sqrt{10}} \begin{bmatrix} 1 & -3 \\ \sqrt{10} & \sqrt{10} \end{bmatrix} \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix}$$

$$v_2^T = \frac{1}{3\sqrt{10}} \begin{bmatrix} -20 & -10 & 20 \\ \sqrt{10} & \sqrt{10} & \sqrt{10} \end{bmatrix}$$

$$v_2^T = \begin{bmatrix} -\frac{20}{30} & -\frac{10}{30} & \frac{20}{30} \end{bmatrix}$$

$$v_2 = \begin{bmatrix} -2/3 \\ -1/3 \\ 2/3 \end{bmatrix}$$

v_3 must be orthogonal to both v_1 and v_2

let v_3 be $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$

$$\begin{array}{l} x + 2y + 2z = 0 \qquad -2x - y + 2z = 0 \\ \begin{bmatrix} 1 & 2 & 2 \\ -2 & -1 & 2 \end{bmatrix} \xrightarrow{R_2 + 2R_1} \begin{bmatrix} 1 & 2 & 2 \\ 0 & 3 & 6 \end{bmatrix} \xrightarrow{R_2 - 2/3 R_1} \begin{bmatrix} 1 & 2 & 2 \\ 0 & 0 & 14/3 \end{bmatrix} \end{array}$$

$$\begin{array}{l} 3 - 2x = 0 \\ 3 = 2x \\ x = 3/2 \end{array} \qquad \begin{array}{l} 6 - 4/3 \\ 18 - 4 \\ 14 \end{array}$$

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$$3y + 6z = 0$$

$$3y = -6z$$

$$y = -2z$$

$$x + 2y + 2z = 0$$

$$x + 2(-2z) + 2z = 0$$

$$\boxed{x = 2z}$$

$$\text{let } z = 1$$

$$x = 2$$

$$y = -2$$

$$\therefore \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2z \\ -2z \\ z \end{pmatrix}$$

$$4z^2 + 4z^2 + z^2 = 1$$

$$9z^2 = 1$$

$$z = 1/3$$

$$\therefore v_3 = \begin{bmatrix} 2/3 \\ -2/3 \\ 1/3 \end{bmatrix}$$

$$\therefore V = \begin{bmatrix} 1/3 & -2/3 & 2/3 \\ 2/3 & -1/3 & -2/3 \\ 2/3 & 2/3 & 1/3 \end{bmatrix}$$

$$\therefore A = U \Sigma V^T$$

$$\begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix} = \frac{1}{\sqrt{10}} \begin{bmatrix} 3 & 1 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} 6\sqrt{10} & 0 & 0 \\ 0 & 3\sqrt{10} & 0 \end{bmatrix} \begin{bmatrix} 1/3 & -2/3 & 2/3 \\ -2/3 & -1/3 & 2/3 \\ 2/3 & 2/3 & 1/3 \end{bmatrix}$$

$$= \frac{1}{\sqrt{10}} \begin{bmatrix} 3 & 1 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} 6\sqrt{10} & 4\sqrt{10} & 4\sqrt{10} \\ -2\sqrt{10} & -\sqrt{10} & 2\sqrt{10} \end{bmatrix}$$

$$A = \begin{bmatrix} 4 & 11 & 14 \\ 8 & 07 & -2 \end{bmatrix} = \frac{1}{\sqrt{10}} \begin{bmatrix} 3 & 1 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} 2\sqrt{10} & 4\sqrt{10} & 4\sqrt{10} \\ -2\sqrt{10} & -\sqrt{10} & 2\sqrt{10} \end{bmatrix} \quad (13)$$

$$= \frac{1}{\sqrt{10}} \begin{bmatrix} 4\sqrt{10} & 11\sqrt{10} & 14\sqrt{10} \\ 8\sqrt{10} & 7\sqrt{10} & -2\sqrt{10} \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix}$$

$$\square \circ \circ \quad A = U \Sigma V^T$$